

Vladimir Dvorkin, Ph.D.

EECS Assistant Professor at the University of Michigan

📍 Ann Arbor, MI ✉ dvorkin@umich.edu 🏆 Scholar 🐦 DvorkinVladimir

APPOINTMENTS & WORK EXPERIENCE	University of Michigan ASSISTANT PROFESSOR DEPARTMENT: ELECTRICAL ENGINEERING AND COMPUTER SCIENCE	Ann Arbor, US 1/2024 - present
	Massachusetts Institute of Technology MSCA-FIBE POSTDOCTORAL FELLOW POSTDOCTORAL ASSOCIATE DEPARTMENT: LABORATORY FOR INFORMATION AND DECISION SYSTEMS & ENERGY INITIATIVE	Cambridge, US 3/2022 - 12/2023 2/2021 - 2/2022
	Georgia Institute of Technology Research Visitor DEPARTMENT: INDUSTRIAL & SYSTEMS ENGINEERING	Atlanta, USA 07/2019 - 12/2019
	Higher School of Economics Research Assistant	Moscow, Russia 12/2013 - 08/2017

EDUCATION	Technical University of Denmark (DTU) PH.D. – ELECTRICAL ENGINEERING M.SC. – SUSTAINABLE ENERGY SUPERVISORS: PROFS. PIERRE PINSON AND JALAL KAZEMPOUR	Lyngby, Denmark 09/2017 - 03/2021 09/2015 - 07/2017
	Higher School of Economics (HSE) M.Sc. – ENERGY ECONOMICS	Moscow, Russia 09/2012 - 06/2014
	Moscow Power Engineering Institute (MPEI) B.Sc. – ELECTRICAL ENGINEERING	Moscow, Russia 09/2008 - 06/2012

AWARDS	🏆 Outstanding Reviewer, IEEE Trans. on Energy Markets, Pol. and Reg.	2024
	🏆 Marie Skłodowska-Curie Actions Postdoctoral Fellowship	03/2022–02/2024
	🏆 Best Paper Award, IEEE Transactions on Power Systems	2021
	🏆 Outstanding Reviewer, IEEE Transactions on Power Systems	2021
	🏆 LANL Grid Science Winter School Scholarship	2019
	🏆 Outstanding Reviewer, IEEE Transactions on Sustainable Energy	2018
	🏆 DTU Tuition Fee Waiver for MSc Students	08/2015–06/2017
	🏆 HSE Scholarship for Science Achievements	2014
	🏆 HSE Scholarship for Excellency	09/2012–06/2014
	🏆 Semifinalist at the Youth Russian Petroleum&Gas Case Championship	2013
	🏆 MPEI Scholarship for Academic Achievements	09/2008–06/2012

FUNDING	1. LearningORDER. (Individual postdoctoral fellowship, 03/2022-02/2024, \$200,000). Awarded by Marie Skłodowska-Curie Actions and Fundación Iberdrola España. Grant agreement No. 101034297. (Executive summary  , presentation )
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PUBLICATIONS

SUBMITTED

- S1. Zhao D., Delikaraoglou S., **Dvorkin**, V., Botterud A., Lamadrid A., 2023. Optimizing bidding curves for renewable energy in two-settlement electricity markets. Submitted to *European Journal of Operational Research*
- S2. **Dvorkin**, V., Fioretto, N., Van Hentenryck, P., Kazempour, J. and Pinson, P., 2024. Privacy-preserving convex optimization: When differential privacy meets stochastic programming. Submitted to *INFORMS Journal on Optimization*
<https://doi.org/10.48550/arXiv.2209.14152>
- S3. Kim, M., **Dvorkin**, V. and Kim, J., 2024. Probabilistic dynamic line rating forecasting with line graph convolutional LSTM. Submitted to *2025 IEEE Power & Energy Society General Meeting (PESGM)*

JOURNAL PUBLICATIONS

- J1. **Dvorkin**, V. 2025. Regression equilibrium in electricity markets. *IEEE Transactions on Energy Markets, Policy and Regulation*, in print.
<https://doi.org/10.48550/arXiv.2405.17753>
- J2. **Dvorkin**, V. 2024. Agent coordination via contextual regression (AgentCONCUR) for data center flexibility. *IEEE Transactions on Power Systems*, in print.
<https://doi.org/10.48550/arXiv.2309.16792>
- J3. Kenis M., **Dvorkin** V., Schittekatte T., Bruninx K., Delarue E., Botterud A., 2023. Evaluating offshore electricity market design considering endogenous infrastructure investments: Zonal or nodal? *IEEE Transactions on Energy Markets, Policy and Regulation*, in print.
- J4. Zhao, D., **Dvorkin**, V., Delikaraoglou, S., Lamadrid, A.J., Botterud, A. 2023. Uncertainty-informed renewable energy scheduling: A scalable bilevel framework. *IEEE Transactions on Energy Markets, Policy and Regulation*, in print.
<https://doi.org/10.1109/TEMPR.2023.3344126>
- J5. **Dvorkin**, V., Botterud, A. 2023. Differentially private algorithms for synthetic power system datasets. *IEEE Control Systems Letters*, vol. 7, pp. 2053-2058
<https://doi.org/10.1109/LCSYS.2023.3284389>
- J6. **Dvorkin**, V., Mallapragada, D. and Botterud, A., 2023. Multi-stage decision rules for power generation & storage investments with performance guarantees. *IEEE Transactions on Power Systems* (in print)
<https://doi.org/10.1109/TPWRS.2023.3257129>
- J7. **Dvorkin**, V., Mallapragada, D., Botterud, A., Kazempour, J. and Pinson, P., 2022. Multi-stage linear decision rules for stochastic control of natural gas networks with linepack. *Electric Power Systems Research (XXII PSCC edition)*, 212, p.108388.
<https://doi.org/10.1016/j.epsr.2022.108388>
- J8. **Dvorkin**, V., Ratha, A., Pinson, P. and Kazempour, J., 2021. Stochastic control and pricing for natural gas networks. *IEEE Transactions on Control of Network Systems*, 9(1), pp.450-462.
<https://doi.org/10.1109/TCNS.2021.3112764>
- J9. **Dvorkin**, V., Fioretto, F., Van Hentenryck, P., Pinson, P. and Kazempour, J., 2021. Differentially private optimal power flow for distribution grids. *IEEE Transactions on Power Systems*, 36(3), pp.2186-2196.
🏆 Best Paper Award for the period 2019–2021
<https://doi.org/10.1109/TPWRS.2020.3031314>
- J10. **Dvorkin**, V., Kazempour, J. and Pinson, P., 2019. Electricity market equilibrium under information asymmetry. *Operations Research Letters*, 47(6), pp.521-526.
<https://doi.org/10.1016/j.orl.2019.09.005>

- J11. **Dvorkin**, V., Delikaraoglou, S. and Morales, J.M., 2018. Setting reserve requirements to approximate the efficiency of the stochastic dispatch. *IEEE Transactions on Power Systems*, 34(2), pp.1524-1536.
<https://doi.org/10.1109/TPWRS.2018.2878723>

CONFERENCE PUBLICATIONS

- C1. **Dvorkin**, V., Fioretto, F. 2023. Price-aware deep learning for electricity markets. *NeurIPS 2023 Workshop on Tackling Climate Change with Machine Learning*
<https://doi.org/10.48550/arXiv.2308.01436>
- C2. **Dvorkin**, V., Chevalier, S., Chatzivasileiadis S., 2023. Emission-constrained optimization of gas networks: Input-convex neural network approach. In *2023 62th IEEE Conference on Decision and Control* (in print).
 ☞ Also selected for as a spotlight talk at 2023 ICLR-CCAI Workshop ☞
<https://doi.org/10.48550/arXiv.2209.08645>
- C3. Zhao, D., **Dvorkin**, V., Delikaraoglou, S., Lamadrid, A. J., Botterud, A., 2023. A scalable bilevel framework for renewable energy scheduling. In *The 14th ACM International Conference on Future Energy Systems (e-Energy) 2023*
<https://doi.org/10.1145/3575813.3595199>
- C4. **Dvorkin**, V., Kazempour, J. and Pinson, P., 2020, August. Chance-constrained equilibrium in electricity markets with asymmetric forecasts. In *2020 International Conference on Probabilistic Methods Applied to Power Systems* (pp. 1-6). IEEE.
 ☞ Best Paper Award Nomination
<https://doi.org/10.1109/PMAPS47429.2020.9183423>
- C5. **Dvorkin**, V., Van Hentenryck, P., Kazempour, J. and Pinson, P., 2020, December. Differentially private distributed optimal power flow. In *2020 59th IEEE Conference on Decision and Control* (pp. 2092-2097). IEEE.
<https://doi.org/10.1109/CDC42340.2020.9303768>
- C6. Radoszynski, A.M., **Dvorkin**, V. and Pinson, P., 2019, June. Accommodating bounded rationality in pricing demand response. In *2019 IEEE Milan PowerTech* (pp. 1-6). IEEE.
<https://doi.org/10.1109/PTC.2019.8810419>
- C7. **Dvorkin**, V., Kazempour, J., Baringo, L. and Pinson, P., 2018, December. A consensus-ADMM approach for strategic generation investment in electricity markets. In *2018 IEEE Conference on Decision and Control* (pp. 780-785). IEEE.
<https://doi.org/10.1109/CDC.2018.8619240>

THESIS

- T1. **Dvorkin**, V., 2021. Stochastic and private energy system optimization. *Ph.D. Thesis*. Technical University of Denmark. (Supervised by Pinson P., Kazempour J. Examined by Chatzivasileiadis, S., Shapiro, A., Wierman, A.)
https://drive.google.com/file/d/1_0wDZ0nnH0tFnDeQ1S-eeW8QYoRJNRa4/view
- T2. **Dvorkin**, V., 2017. Multi-stage strategic investment in CCGTs and wind power units via progressive hedging. *M.Sc. Thesis*. Technical University of Denmark. (Supervised by Pinson P., Kazempour J. Examined by Boomsma, T.K.)
<https://drive.google.com/file/d/16MFeiUVbQ4IQ-d6wvUF9jZYUU-RHucYa/view>

COURSES

- | | |
|---|-------------|
| 1. <i>EECS 598: Computational power systems</i> | Winter 2025 |
| 2. <i>EECS 463: Power system design and operation</i> | Fall 2024 |
| 3. <i>EECS 559: Optimization methods for SIPML</i> | Winter 2024 |
| 4. <i>Renewables in Electricity Markets</i> | DTU |
| Head teaching assistant | Spring 2020 |
| Teaching assistant | Spring 2017 |

	5. <i>DTU Summer School on Energy Optimization, Learning and Game Theory</i>	DTU Teaching assistant Summer 2017–2019
	6. <i>Advanced Optimization in Electricity Markets</i>	DTU Teaching assistant Fall 2018
	7. <i>Decomposition Techniques for Energy Systems Applications</i>	Skoltech Teaching assistant, lecturer Fall 2018
PH.D. STUDENTS	1. Xinwei Liu	starting Fall 2024
	2. Shengyang Wu	starting Fall 2024
	3. Milad Hoseinpour Valoujaei	starting Fall 2024
M.SC. STUDENTS	1. Xinwei Liu, Decision-focused post-processing of trained neural networks.	Winter 2024, UMich.
	2. Hyun June Kim, Synthesizing data for linear programming.	Winter 2024, UMich.
	3. Samuel A. Abel, Fueling an energy transition: Designing an optimal portfolio of competing fuels under uncertainty.	Spring 2023, MIT.
	4. Greta Marija Nikkare, Co-optimization of green hydrogen and power system expansion planning.	Spring 2022, MIT.
	5. Rafal Michal Mikulowski, Power systems operation and planning using chance-constrained programming.	Fall 2019, DTU.
	6. Andrea Marin Radoszynski, Demand response and bounded rationality in electricity markets.	Spring 2018, DTU.
	7. Eirini Ioanna Barmpati, Stochastic equilibrium models for capacity investment in energy systems.	Spring 2018, DTU.
UROP STUDENTS	1. Shreya Chaudhary, Synthetic Energy Dataset Generation.	Winter 2023, MIT
VISITING PH.D. STUDENTS	1. Philipp Gunkel , Renewable energy trajectory of no-regret.	Fall 2023, MIT.
	2. Michiel Kenis, Toward off-shore bidding zones: the role of generation and transmission capacity investments.	Fall 2022, MIT.
PHD THESIS COMMITTEES	1. Hannah Moring, University of Michigan – Ann Arbor	2024
	2. Sunny Chen, University of Michigan – Ann Arbor	2024
	3. Austin Lin, University of Michigan – Ann Arbor	2024
TEACHING TRAINING	1. MIT Kaufman Teaching Certificate Program (description ↗).	Fall 2022.

SELECTED INVITED TALKS	1. <i>Data center–power grid coordination under engineering and privacy constraints</i> Boston University (CISE Seminar)	November, 2024.
	2. <i>Regression Nash equilibrium in electricity markets</i> University of Michigan (Control Seminars)	November, 2024.
	3. <i>Carbon-aware computing: How to get power systems and data centers to talk</i> 2024 IEEE Control Systems Society Day	October, 2024.
	4. <i>Formal privacy guarantees for optimization datasets in power systems</i> University of Michigan (Comm. & Signal Processing Seminar)	March, 2024.
	5. <i>Privacy-preserving optimization and learning via stochastic optimization</i> Georgia Institute of Technology (AI4OPT)	March, 2023.
	6. <i>Optimization and learning in energy systems: Privacy and performance.</i> University of Michigan (ECE Department)	May, 2023.
	Massachusetts Institute of Technology (CEE Department)	February, 2023.
	University of Wisconsin–Madison (ECE Department)	February, 2023.
	University of Minnesota (ISyE Department)	January, 2023.
	University of Edinburgh (School of Mathematics)	December, 2022.
	7. <i>Differential privacy meets stochastic programming.</i> Copenhagen University (Department of Computer Science). Hosted by: YEVGENY SELDIN	November, 2022.
	8. <i>Performance guarantees for investments in power systems under uncertainty.</i> Technical University of Denmark (DTU Management). Presented at: SEMINAR ON ECONOMICS OF GREEN TRANSITION	November, 2022.
	9. <i>Privacy-preserving perturbation of convex optimization programs.</i> California Institute of Technology. Hosted by: ADAM WIERMAN and STEVEN LOW	August, 2022.
	10. <i>Privacy-preserving perturbation of convex optimization programs.</i> Massachusetts Institute of Technology. Presented at STATS&LIDS TEA TALKS seminar series	May, 2022.
CONFERENCES & WORKSHOPS	11. <i>Algorithmic privacy for energy system optimization.</i> Massachusetts Institute of Technology. Presented at MITEI RESEARCH MEETS seminar series	May, 2022.
	12. <i>Stochastic control and market design for natural gas networks.</i> Massachusetts Institute of Technology. Hosted by: AUDUN BOTTERUD	September, 2020.
	13. <i>Differentially private optimization of power systems.</i> Georgia Institute of Technology. Presented at DOS SEMINARS seminar series	December, 2019.
	14. <i>Electricity market equilibrium under information asymmetry.</i> Johns Hopkins University. Hosted by: BENJAMIN HOBBS	January, 2019.
	1. <i>US electric grid decarbonization pathways under market and policy uncertainties</i> Workshop at National Academies (poster)	September, 2024.
	2. <i>Regression Nash equilibrium in electricity markets</i> 2024 INFORMS Annual Meeting, Seattle	October, 2024
	3. <i>Machine learning in electricity markets: Is there equilibrium?</i> Institute for Mathematical and Statistical Innovation, Chicago	August, 2024
	4. <i>Equilibrium forecasting in electricity markets</i> 2024 IEEE Power & Energy Society General Meeting	July, 2024

5. *Trustworthy deep learning for electricity market applications*
Federal Energy Regulatory Commission July, 2024
6. *Forecast equilibrium in electricity markets*
33rd European Conference on Operational Research July, 2024
7. *Price-aware forecasting in electricity markets*
Institute for Mathematical and Statistical Innovation, Chicago June, 2024
8. *Privacy-preserving synthetic dataset generation for power systems research*
2023 INFORMS Annual Meeting. October, 2023
Federal Energy Regulatory Commission. July, 2023
9. *Emission-constrained optimization of gas systems with input-convex neural nets*
ICLR Workshop: Tackling Climate Change with Machine Learning May, 2023
10. *Privacy-preserving machine learning by means of stochastic optimization.*
2023 MLTea talks at MIT February, 2023
11. *Algorithmic privacy for energy systems optimization.*
2022 INFORMS Annual Meeting. October, 2022
12. *Multi-stage stochastic generation investment with performance guarantees.*
MITEI Future Energy Systems Center Fall 2021 Workshop. December, 2021
13. *Multi-stage investment decision rules for power systems: sensitivities, deterministic equivalents, and performance guarantees.*
2021 INFORMS Annual Meeting. October, 2021
14. *Multi-stage stochastic generation investment with performance guarantees.*
Federal Energy Regulatory Commission. June, 2021
15. *Differentially private optimal power flow for distribution grids.*
IEEE PES Madrid PowerTech 2021. June, 2021
16. *Stochastic control and market design for natural gas networks.*
2020 INFORMS Annual Meeting. October, 2020
17. *Differentially private optimal power flow for distribution grids.*
2020 INFORMS Annual Meeting. October, 2020
18. *Differentially private distributed optimal power flow.*
2019 GeorgiaTech Energy Systems and Optimization Workshop. November, 2019
19. *Electricity market equilibrium under information asymmetry.*
2019 INFORMS Annual Meeting. October, 2019
20. *Electricity market equilibrium under information asymmetry.*
2019 IEEE PES General Meeting. August, 2019
21. *Electricity market equilibrium under information asymmetry.*
XV International Conference on Stochastic Programming. August, 2019
22. *Power system optimization under information asymmetry.*
Grid Science Winter School, Los Alamos National Laboratory. January, 2019
23. *Consensus-ADMM approach for strategic investment in electricity markets.*
2018 IEEE Conference on Decision and Control. December, 2018
24. *A solution framework for strategic investment problems via progressive hedging.*
XV Conference on Computational Management Science. May, 2018

REVIEWER SERVICE	1. ICLR Tackling Climate Change with Machine Learning Workshop 2. IEEE Transactions on Smart Grids 3. IEEE Transactions on Automatic Control 4. IEEE Transactions on Sustainable Energy 5. IEEE Transactions on Power Systems 6. European Journal of Operational Research 7. International Transactions on Electrical Energy Systems 8. PSCC – Power Systems Computation Conference 9. IEEE Conference on Decision and Control 10. IEEE PES PowerTech 11. IEEE American Control Conference	since 2024 since 2019 since 2019 since 2018 since 2018 since 2020 since 2017 since 2018 since 2018 since 2019 since 2018
GITHUB REPOSITORIES	1. PrivateOpt: Differentially Private Convex Optimization ↗ 2. InvestmentLDR: Investment Linear Decision Rules for Power Systems ↗ 3. DP-CC-OPF: Differentially Private Chance-Constrained OPF ↗ 4. GasLDR: Linear Decision Rules for Stochastic Control of Gas Networks ↗ 5. Stochastic Control and Pricing for Natural Gas Networks ↗	
PROFESSIONAL MEMBERSHIPS	IEEE, Member (Power and Energy Society) since 2017 INFORMS, Member (Energy, Natural Resources and Environment section) since 2019.	
OTHER	1. Founder of the ENOPTIMAL: ENERGY, OPTIMIZATION AND LEARNING ↗ seminar series at MIT to bridge energy researchers during the pandemic.	